S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	SUB TOTAL OF (I)- FACTORY PEB WORKS			L.S.	
II	SUPPLYING AND FIXING OF FACTORY JOINERY WORKS (SUCH AS DOORS, WINDOWS, ROLLING SHUTTERS ETC)				
1 (a)	**Aluminium fixed cum top hung openable windows**				
	Providing and fixing **aluminium fixed cum top hung openable windows of Troptec AS 39 of Schuco or 'E' series of Kawneer or Topaz of Technal make** or approved equivalent make and fabricate the windows as per drawing and fixed at all levels, elevations and heights. All aluminium sections shall be satisfied the design wind pressure (Minimum wind pressure shall be 200 Kg/Sqm) as per IS code or relevant international code. All aluminium sections shall be 63400 (H9) grade conforming to IS 8147 and shall be finished with approved colour anodising of thickness not less than 25 microns.				
	Main / outer frames shall be fixed into masonry wall / RC concrete surface with necessary clamps, fasteners with Hilti make, nuts, bolts, rivets, washers and other fastening materials shall be in stainless steel SS 316 grade. Rate shall include the cost for providing and fixing of aluminium 'L' angle - 2.0mm thick with suitable size as per site and aluminium flashing 1.2 mm thick finished with approved colour anodising of thickness not less than 25 microns with suitable size as per site.				
	Rate shall include for designing, providing and fixing hardwares for top hung openable outward shutter having Multi point lock with required points based on the window panel size and handle with key of Model no 610 K or Alan key type model of Alualpha make, SS friction hinges of Model no SPT 26, Corner Transmission Model no 9312, corner locking point model no 9313 and restrictor stay arm assembly model no SR 8 (Opening with restrict to 200mm from the frame plan) with release key of model no R 7437 of Securestyle (UK) make or approved equivalent make				
	Window shutter shall be provided with Insulated glass of 24 mm thick having 6 mm thick Reflective coated low E glass on # 2 surface - **KT 740 Blue of Saint Gobain make or Blue Vision LH 24/20 of Asahi make** as outer pane and 12 mm airgap and 6 mm thick glass uncoated clear as inner pane of same make and both panes are heat strengthened. Rate shall include the cost for scaffolding, non staining weather silicon sealant of Dowcorning or approved equivalent make shall be provided at the junctions between wall, main frames and wherever required, stainless steel safety clips at each corner of glass as an additional safety to avoid breakage of corner of the glass and also to be provided with EPDM gasket as weather strip to ensure water and air tightness and shall include providing necessary hardwares, concealed locks and Handle etc. complete all as directed.				
	Size -4500 x 1500	Nos	8		

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
1(b)	**Aluminium fixed cum sliding windows**				
	Providing and fixing **aluminium fixed cum sliding windows of Troptec AS 39 of Schuco or 'E' series of Kawneer or Topaz of Technal make** or approved equivalent make and fabricate the windows as per drawing and fixed at all levels, elevations and heights. All aluminium sections shall be satisfied the design wind pressure (Minimum wind pressure shall be 200 Kg/Sqm) as per IS code or relevant international code. All aluminium sections shall be 63400 (H9) grade conforming to IS 8147 and shall be finished with approved colour anodising of thickness not less than 25 microns.				
	Main / outer frames shall be fixed into masonry wall / RC concrete surface with necessary clamps, fasteners with Hilti make, nuts, bolts, rivets, washers and other fastening materials shall be in stainless steel SS 316 grade. Rate shall include the cost for providing and fixing of aluminium 'L' angle - 2.0mm thick with suitable size as per site and aluminium flashing 1.2 mm thick finished with approved colour anodising of thickness not less than 25 microns with suitable size as per site.				
	Window shutter shall be provided with Insulated glass of 24 mm thick having 6 mm thick Reflective coated low E glass on # 2 surface - **KT 740 Blue of Saint Gobain make or Blue Vision LH 24/20 of Asahi make** as outer pane and 12 mm airgap and 6 mm thick glass uncoated clear as inner pane of same make and both panes are heat strengthened. Rate shall include the cost for scaffolding, non staining weather silicon sealant of Dowcorning or approved equivalent make shall be provided at the junctions between wall, main frames and wherever required, stainless steel safety clips at each corner of glass as an additional safety to avoid breakage of corner of the glass and also to be provided with EPDM gasket as weather strip to ensure water and air tightness and shall include providing necessary hardwares, concealed locks and Handle etc. complete all as directed.				
	Size -4500 x 1500	Nos	8		
1 (c)	Removing of the exisiting Aluminium windows with frames, shutter and with all fittings along Grid 'J' of phase-1 expansion side and refixing the same along the Grid 'S' of phase 2 and phase 3 expansion side including removal of debris, disposing the same away from the site, loading and unloading, making the surface good etc. complete. Rate shall include for necessary tools and plants, scaffolding, water and power charges, cleaning the surface, preparation of surface. Rate shall include for the removed materials to be stored properly /carefully before refixing in the relevant location				
	Size -4500 x 1500	Nos	8		
2 (a)	**Single / Double leaf door - Regular/1 Hr Fire escape steel door**				

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	Providing and fixing of Hollow metal door at all levels from ISO 9001-2000 certified Manufacturer. All hollow metal general doors with or without vision panel. Pressed Galvanized steel Single /Double leaf to required sizes and approved make which consists of frame, shutter, infill , necessary hardwares and finish as detailed below. Doors shall be supplied as per the drawing.				
	Door frame shall be Single rebate profile of size 100 x 57 mm made out of 1.20mm thick galvanised steel sheet (18 gauge). Frames shall be Mitered and field assembled with self tabs. All provision should be mortised, drilled and tapped for receiving appropriate hardware. Rubber door silencers should be provided on the striking jamb. Frames should be provided with back plate bracket and anchor fasteners for installation on a finished plastered masonry wall opening. Once frame installed should be grouted with cement & sand slurry necessary for fire doors on the clear masonry opening.Grouting holes on the frame shall be sealed with rubber cap of adequate size and thickness.				
	Door leaf should be 46mm thick fully flush double skin door with or without vision lite. Door leaf shall be manufactured from 0.8mm (22 guage) minimum thick galvanised steel sheet. Inverted Verticle steel stiffeners to be provided at a distance of 6 inches, the gaps shall be filled with mineral wool or rock wool.or any proprietary insulation material. All doors should be factory prepped for receiving appropriate hardware and provided with necessary reinforcement for hinges, locks, and door closers as per standard. The edges should be interlocked with a bending radius of 1.4mm. For pair of doors astragals has to be provided on the meeting stile for both active and inactive leaf.				
	All doors and frames shall be finished with etched primer coating, stove zinc phosphate primer and thermosetting **polyurethane** aliphatic grade **paint** or exterior grade water based Polyster powder coating not less than DFT 65 microns of Corro-Coat PE 7 series of JOTUN make of approved colour. The door leaf and frame shall have passed minimum 250 hours of salt sprey test.and specified fire rating after application and painting. And ensure that the paint is having non combustiable properties. Door frame and shutter shall have reinforcement back plate of 3mm minimum thickness to receive hardwares and to receive required bite thickness for screws and bolts.				
	Rate shall include supply and installation of door and hardwares as mentioned below and also confirms as per drawing for the specified makes and models.				
	HINGES: Stainless Steel Ball Bearing Butt Hinges 3mm Thick and shall be Fixed Flush To The frame & Shutter.(Minimum 3 hinges per shutter for width below 1100mm and height 2100mm and 4 hinges per shutter for width more than 1100mm and height more than 2100mm)				

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	LOCK: Mortise sash lock with Lever handles, Mortise dead bolt, Mortise latch and panic devices etc can be provided as required.				
	1 No.Door closer of heavy duty EN 2-6 with standard arm with PA bracket in active leaf				
	FLUSH BOLTS (DOUBLE DOOR) Concealed extended lever action flush bolts provide on the top and bottom of the door to the leading stile edge.				
	SMOKE SEAL Self adhesive sponge seal for frames in Black colour - on 3 sides to suit both Single/Double Leaf shutter of 8144 S or 188 S of Guardian or equivalent				
	DOOR SEAL Door Sweep Bottom Brush Seal with colour anodized aluminium frame to suit both Single/Double Leaf of 8192 AA or 8193 AA of Guardian or equivalent				
	Vision panel wherever applicable should be of size : 200x600mm for each shutter provided as per manufacturers recommendation with a clip on arrangement. The glass should be 6mm thick clear borosilicate fire rated glass of equal rating of "Pyron" or st.Gobain make.				
	Rim panic latch with outside entry function of approved model and make (Function 09 - Lockable Lever) suitable for single /Double leaf door with one side Key operation & SS Handle of approved make & model instead of Mortise sash lock / Dead lock with both side key operation				
	Single leaf steel doors with all specifications and hardwares as mentioned above and confirming to the drawing with the following classification:				
(i)	Single leaf door steel door with panic bar and vision panel - Size 1.2 x2.5m	Nos	3.00		
2(b)	Removing of the exisiting steel doors with frames, shutters and including all hardwares along Grid 'J' of phase-1 expansion side and refixing the same along the Grid 'S' of phase 2 and phase 3 expansion side including removal of debris, disposing the same away from the site, loading and unloading, making the surface good etc. complete. Rate shall include for necessary tools and plants, scaffolding, water and power charges, cleaning the surface, preparation of surface.Rate shall include for the removed materials to be stored properly /carefully before refixing in the relevant location				
(i)	Single leaf door steel door with panic bar and vision panel - Size 1.2 x2.5m	Nos	2.00		
3	**Single / Double leaf door - 2 Hr fire rating Fire Escape Door**				

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	Providing and fixing of Hollow metal fire door at all levels from ISO 9001-2000 certified Manufacturer. Fire door should be as per NFPA 80 and NFPA 252. All fire doors should be tested at UL for maximum rating of 2Hrs and should pass both UL10B and UL10C. Pressed Galvanized steel Single /Double leaf to required sizes for approved rating and approved make which consists of frame, shutter, infill, necessary hardwares and finish as detailed below. Doors shall be supplied with appropriate labels confirming the rating.				
	Door frame shall be double rebate profile of size 143 x 57mm made out of 1.60mm (16gauge) minimum thick galvanized steel sheet. Frames shall be Mitered and field assembled with self tabs. All provision should be mortised, drilled and tapped for receiving appropriate UL labled hardware. Rubber door silencers should be provided on the striking jamb. Frames should be provided with back plate bracket and anchor fasteners for installation on a finished plastered masonry wall opening. Once frame installed should be grouted with cement & sand slurry necessary for fire doors on the clear masonry opening. Grouting holes on the frame shall be sealed with rubber cap of adequate size and thickness.				
	Door leaf shall be 46mm thick fully flush double skin door with or without vision lite. Door leaf shall be manufactured from 1.2mm (18guage) minimum thick galvanised steel sheet. Inverted Verticle steel stiffeners to be provided at a distance of 6 inches, the gaps shall be filled with mineral wool or rock wool or any proprietary insulation material satisfying the fire rating norms and also satisfying FM / UL label requirements.All doors should be factory prepped for receiving appropriate hardware and provided with necessary reinforcement for hinges, locks, and door closers as per standard. The edges should be interlocked with a bending radius of 1.4mm. For pair of doors astragals has to be provided on the meeting stile for both active and inactive leaf.				
	All doors and frames shall be finished with etched primer coating, stove zinc phosphate primer and thermosetting **polyurethane** aliphatic grade **paint** or exterior grade water based Polyster powder coating not less than DFT 65 microns of Corro-Coat PE 7 series of JOTUN make of approved colour. The door leaf and frame shall have passed minimum 250 hours of salt sprey test.and specified fire rating after application and painting. And ensure that the paint is having non combustiable properties. Door frame and shutter shall have reinforcement back plate of 3mm minimum thickness to receive hardwares and to receive required bite thickness for screws and bolts.				
	Rate shall include supply and installation of door and hardwares as mentioned below and also confirms as per drawing for the specified makes and models.				

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	HINGES: Stainless Steel Ball Bearing Butt Hinges 3mm Thick and shall be Fixed Flush To The frame & Shutter.(Minimum 3 hinges per shutter for width below 1100mm and height 2100mm and 4 hinges per shutter for width more than 1100mm and height more than 2100mm)				
	LOCK: Mortise sash lock with Lever handles, Mortise dead bolt, Mortise latch and panic devices etc can be provided as required.				
	1 No.Door closer of heavy duty EN 2-6 with standard arm with PA bracket in active leaf				
	FLUSH BOLTS (DOUBLE DOOR) Concealed extended lever action flush bolts provide on the top and bottom of the door to the leading stile edge.				
	SMOKE SEAL Self adhesive sponge seal for frames in Black colour - on 3 sides to suit both Single/Double Leaf shutter of 8144 S or 188 S of Guardian or equivalent				
	DOOR SEAL Door Sweep Bottom Brush Seal with colour anodized aluminium frame to suit both Single/Double Leaf of 8192 AA or 8193 AA of Guardian or equivalent				
	Vision panel wherever applicable should be of size : 200x600mm for each shutter provided as per manufacturers recommendation with a clip on arrangement. The glass should be 6mm thick clear borosilicate fire rated glass of equal rating of "Pyron" or st.Gobain make.				
	Rim panic latch with outside entry function of approved model and make (Function 09 - Lockable Lever) suitable for single /Double leaf door with one side Key operation & SS Handle of approved make & model instead of Mortise sash lock / Dead lock with both side key operation				
	NOTE : Necessary CBRI certificate shall be produced for the steel door to confirm the fire rating specified in the specification and necessary UL label to be produced for the shutter.				
	Single leaf 2 Hr fire rating doors with all specifications and hardwares as mentioned above and confirming to the drawing with the following classification:				
(i)	Single leaf door 2 Hr fire rated door with panic bar and vision panel - Size 1.2 x2.5m	Nos	1.00		
4	**High speed sectional door**				

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	Providing and fixing of High speed sectional doors of Door leaf (door sections made of steel panels). Door sections made of double-skinned, PU-foamed steel panels, 100 % CFC-free, outside and inside stucco-textured (made of hot galvanized steel)625/750 mm high (max. door width 5000 mm) or 500/375 mm high, depth 42 mm, evenly spaced increments of 125 mm.				
	Item characteristics to EN 13241-1 EN 12424 Resistance to wind load Class 3 EN 12425 Watertightness (Details for door with and without wicket door) Class 3 EN 12426 Air permeability (Details for door without wicket door) Class 2 Air permeability (Details for door with wicket door) Class 1 EN 13241, Thermal insulation U= 1.0 W/m2K Appendix B (Details for door without wicket door: 1.0 EN 12428 approx. 25 m2 door area) Thermal insulation U= W/m2K (Details for door with wicket door: 1.2 approx. 25 m2 door area) EN 717-1 Acoustic-rating value R = 25dB (Details for door without wicket door) 22 Frame/track application				
	Laterally closed, profiled angle frame, made of galvanised steel, with safety track secured with screws. Door lock Manually operated: Inside locking with shootbolt, rotary latch (on request for track applications with low-mounted torsion spring shaft) or floor locking included in the door price. Power-driven: Inside locking with shootbolt at a surcharge. Counterbalance Torsions spring, lateral carrying cable (for low headroom track combination made of carrying chain and cable). Safety features acc. to DIN EN 12604				

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	• Manually operated doors with one torsion spring with approved catch safety device 1) • Manually operated doors with more than one torsion spring with approved spring safety device 1), doors higher than 5000 mm approved catch safety device on both sides 1) (not for DPU) • Finger trap protection outside and inside 1) (not for DPU) 1) European Patent Seals Bottom seal made of 3-chamber EPDM profile with flexible adjustment lip, side seal, lintel seal, intermediate seals between the sections. Operator: With 'Frequency Invertor control with soft start and soft stop, with intermediate half open button.				
	Surface protection Door leaf polyester-primer coated, outside and inside grey white, based on RAL 9002. Frames with glazing in anodized aluminium extension in natural finish E6/EV1 to DIN 17611. Polyester-primer coating in preferred colours inside grey white, based on RAL 9002 (Coil-Coating) Outside in preferred colour see right, based on RAL 1021 Rape yellow RAL 7016 Anthracite grey / RAL 3000 Fire red RAL 8028 Terra brown / RAL 5002 Ultramarine blue RAL 9002 Grey white / RAL 5009 Azure blue RAL 9006 White aluminium / RAL 5010 Gentian blue RAL 9007 Grey aluminium / RAL 6002 Leaf green RAL 9010 Pure white / RAL 6005 Moss green RAL 9016 Traffic white				
	Generelly not coated: • Complete door frame Door leaf reinforcements Anodized wicket door profile Clip profiles of aluminium frames Infill in stainless steel perforated sheet End caps of door sections All other steel parts are made of galvanised steel. Coated torsion spring.				
	Operation: Manual Operation with Hand Chain Hoist				
	Operator & Control for				
	Operator WA 400 with control A445:				

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	Complete ready-to-fit shaft operator with cable slack device Power input (3-phase current: 0.37 kW) Protection category IP 65 Electronic absolute encoder (AWG) for determining door position				
	Control unit A 445:				
	Microprocessor control in separate housing Open-stop-Close membrane push button integrated in housing Miniature lock Adjustable force limit Protection category IP 65 Self monitoring closing edge safety device (SKS) with optosensors Connecting cable with CEE plug in protection category IP44 Fitting of the control immediately next to sectional door, plug in door leaf sensors.				
	Vision Panel / Glazing: To be provided				
a)	HSSD1-Size -3500mm x4500mm	Nos	2		
5	**High speed Rolling shutter**				
	Supply & Installation of **Hörmann Motorised Rolling Shutter** Type Decotherm S SB, made of double skinned insulated panels with CFC Free PU Foam, for use in Industries:				
	Door Leaf:				
	Door made of double-skinned, PU-foamed steel sections				
	100% CFC-free & made of hot galvanized steel				
	Profile height of 109mm.				
	Performance characteristics:				
	EN 12424 - Resistance to wind: Class 2 (100km/hr)				
	EN 13241 - Thermal insulation U = 3.9 W/m^2K				
	EN 717 - 1 - Acoustic Insulation R = 19 dB				
	Side Guide:				
	Enclosed, moulded angle frame, made of aluminium secured with screws.				
	Black plastic runner with lintel seals				
	Safety features according to 13241 - 1:				

S.No	Description	Qty	Unit	Rate	Amount (Rs.)
	Manual operation using a tension spring assembly with approved catch safety device.				
	Safety features in accordance to DIN en 13241-1 shall be followed.				
	Surface Protection:				
	- Door leaf coated with protective paint				
	- Steel Parts manufactured from hot galvanized steel.				
	Colors will be chosen from the below for Outside surface:				
	RAL 7016 - Anthracite grey				
	RAL 3000 - Flame Red, RAL 5009 - Azure Blue				
	RAL 9002 - Grey white, RAL 6002 - Leef Green				
	RAL 9006 - White aluminium				
	RAL 5010 - Genitan blue, RAL 9007 - Grey aluminium				
	RAL 8028 - Terra Brown				
	Counterbalance: Tension springs along the side guides				
	Operator WA 300 R S4				
	- Standard soft start & soft stop for gentle & quiet travel				
	- Power input (Single phase 230V)				
	- Operating Temperature from -20 degrees C to 60 degrees C				
	- Protection category IP 65				
	- Standard maintenance release				
	Control unit DTH R:				
	- Microprocessor control in separate housing				
	- Open-stop-Close membrane push button integrated in housing				
	- Protection category IP 65				
	- Self monitoring closing edge safety device with optosensors.				
	- Connecting cable with CEE plug in protection category IP44				
	- Fitting of the control immediately next to rolling shutter				
	Vision Panel : Glazing to be provided for 8 profiles				
	Note: Additional stiffners may be considered for height more than 3m in PEB scope.				
	Clear opening size:				
(i)	HSRS1-Size -3500mm x6000mm	Nos	4.00		
	SUB TOTAL OF (II)- SUPPLYING AND FIXING OF FACTORY JOINERY WORKS				